

HexaHeal: how long can a six-legged robot wait before it reacts to a lost leg?

Technical note, v4. Simulation only (MuJoCo, 2 vCPU sandbox). No hardware.

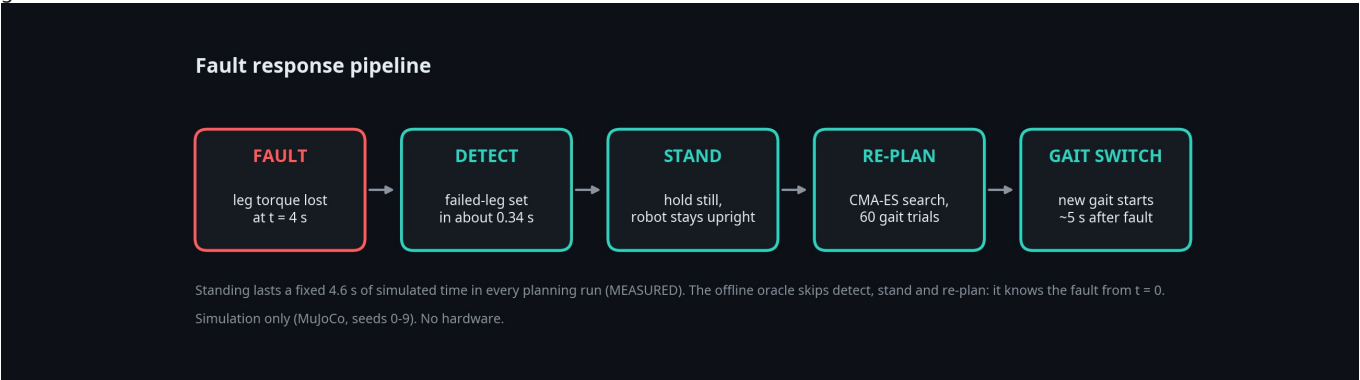
Limits. Simulation only; disabled-leg faults on flat ground, 14 s episodes, fault at 4 s, 0.25 m/s command. 10 evaluation seeds (0-9); tuning used seeds 100-109 only. The oracle is an offline upper bound, not a controller. 9 of 21 cases are recovered by no controller; that is not proof they are impossible. Sim-to-real transfer is UNKNOWN.

Question and metric

When 1 or 2 of 6 legs fail (21 cases), in how many does a controller **recover**? *Upright* = no fall. *Recovered* = no fall and at least 0.125 m/s over the last 8 s; a case counts if it holds on at least 7 of 10 seeds. An earlier metric scored a robot that stood still as a success: all 75 upright runs of the plain tripod were standing still.

Controller

The controller finds the failed-leg set in about 0.34 s (mean, MEASURED), stands, then searches a new free gait online with CMA-ES (60 trials over per-leg phase, duty, amplitude, lift) and switches. The stand lasts a fixed 4.6 s of simulated time in every planning run (MEASURED), so the new gait starts about 5 s after the fault.



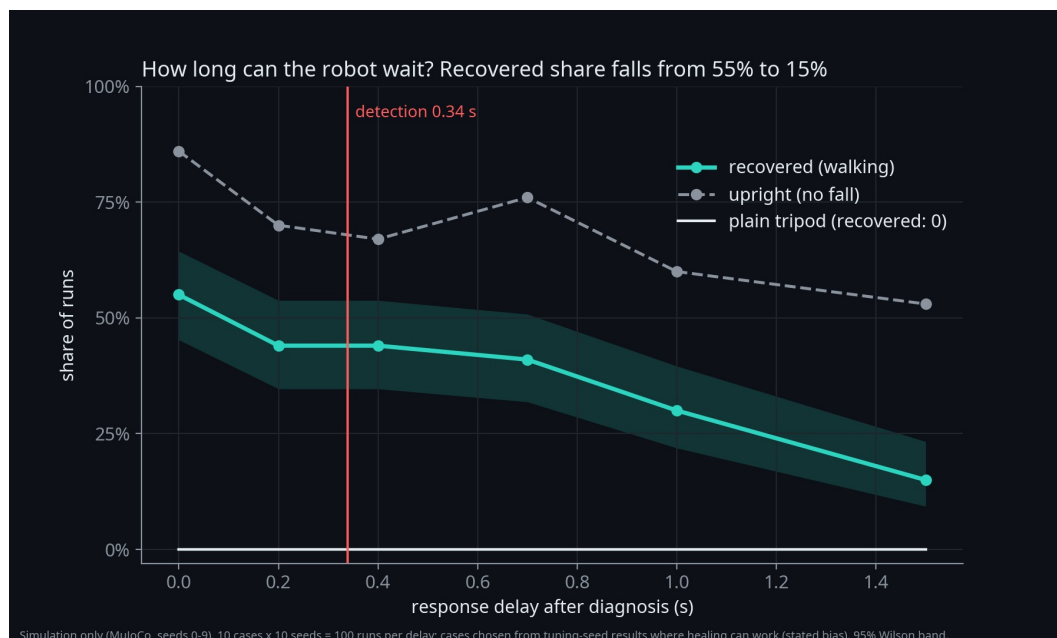
Headline (MEASURED, seeds 0-9)

controller	recovered /21	upright /21	upright runs standing still /210
plain tripod	0	6	75
online re-planning (Tier B)	6	12	54
Tier B+ warm start (not adopted)	9	14	39
oracle (offline, knows fault)	15	18	26

Tier B minus plain tripod: +6 recovered cases, 95% paired-bootstrap CI [+4, +8]. **Warm start: no gain by the pre-registered rule** (+3, CI [0, +6]; the rule needed at least +2 and an interval that excludes zero).

Latency

Delaying the repair after diagnosis: recovered share 55% at 0 s, 44% at 0.2 s, 41% at 0.7 s, 30% at 1.0 s, 15% at 1.5 s (100 runs per delay). Front-leg faults (R1, L1) fall to 0/10 recovered at 0.2 s. *Stated bias*: the 10 cases were chosen from tuning-seed results as cases where healing can work.



Per case

Recovered cases: 21 fault cases x 10 seeds

	R1	R2	R3	L1	L2	L3	R1+R2	R1+R3	R1+L1	R1+L2	R1+L3	R2+R3	R2+L1	R2+L2	R2+L3	R3+L1	R3+L2	R3+L3	L1+L2	L1+L3	L2+L3
oracle (knows fault) (15/21)	10	8	10	10	9	10	1	6	5	8	10	0	10	10	10	10	10	9	4	10	4
final controller (Tier B) (6/21)	3	8	5	6	8	9	0	0	0	1	5	0	8	2	7	0	7	0	0	0	0
Tier B+ (not adopted) (9/21)	5	9	7	10	10	10	0	0	0	4	9	0	10	0	8	0	8	0	0	0	0
plain tripod (0/21)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Cell = recovered runs of 10 seeds. Teal: recovered in at least 7 of 10. Dark: upright, not recovered. Red: falls in more than 3 of 10.

Simulation only (MuJoCo, seeds 0-9). Recovered = no fall and at least 0.125 m/s over the last 8 s. Oracle row: fault from t=0, 15 s run (different conditions).

What failed or did not help

- Healing v1: 0 recovered, fewer upright cases than the plain tripod (4 vs 6).
- Two tuning rounds (do-no-harm rule, steady-state fitness) lowered recovery on tuning seeds; the config was left unchanged.
- Warm start: no gain by the rule. The Tier B stand-still is a fixed budget; most of its upright runs that end still are not walking after the switch (diagnostic in docs/RESULTS_V4.md). Whether a shorter stand would help is UNTESTED.
- PPO reference: 1 recovered case. Connectome-inspired controller: a small MLP beat it; archived.

Reproduce

make reproduce-v4 rebuilds stats and figures from committed results; bash scripts/reproduce_v4.sh full re-runs the episodes (deterministic). Pre-registrations: docs/PREREGISTRATION_V3.md, V4. Claims register: docs/RESULTS_V3.md, RESULTS_V4.md.